

Homework 2

Exponentials and logarithms · about 60 minutes

Each question graded N0–E8

Show all working. Give answers to 3 decimal places unless the question asks for an exact answer. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

1 Write $5^3 = 125$ in log form. [A]

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2 Mia says $\log(10 + 10) = \log 10 + \log 10 = 2$. Is she right? Give a reason. [A]

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3 Write x^{-2} without a negative index. [A]

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4 Factorise $x^2 - 49$. [A]

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QUESTION ONE

(a) Evaluate (i) $\log_2 128$ (ii) $\log_{25} 5$ [A]

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(b) Solve $\log_x 343 = 3$. [A]

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(c) Write $2 \log(3a) - \log\left(\frac{a}{4}\right)$ as a single logarithm in its simplest form. [M]

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(d) Solve $\log_4(x - 1) + \log_4(x + 5) = 2$. [M]

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- (e) $\log_2 x + \log_2 y = 3$ and $\log_2 x - \log_2 y = 1$. Find x and y , explaining your reasoning. [E]

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QUESTION TWO

- (a) Solve $3^x = 40$. [A]

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- (b) Solve $2^{x+3} = 100$. [A]

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- (c) Solve $25^x = 125^{x-2}$ exactly. [M]

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- (d) Sam solved $4^{2x+1} = 60$. **Sam's answer has 2 errors.** Find each one, fix it, and say why it loses marks. [M]

Sam's answer

$$2x + 1 \log 4 = \log 60$$

$$2x + 1 = \log 15 = 1.176$$

$$x = 0.088$$

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- (e) Solve $9^x - 10 \times 3^x + 9 = 0$. [E]

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QUESTION THREE

A town of 8000 people grows by 3% each year, so its population is $P = 8000(1.03)^t$ after t years.

- (a) Find the population after 5 years. **[A]**

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- (b) When will the population first reach 12 000? **[A]**

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- (c) A 400 mg dose of a medicine decays so that 250 mg remain after 6 hours. Find the percentage lost each hour. **[M]**

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- (d) A kettle's water cools as $T = 22 + 78e^{-0.05t}$ (t in minutes). When is it 40°C ? **[M]**

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- (e) Account A holds \$5000 growing at 4% a year; account B holds \$4000 growing at 6% a year. After how many whole years does B first hold more than A? Show your reasoning with logs. **[E]**

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